

Deep Learning Based ECG Signal Classification

Abstract

The classification of Electrocardiogram (ECG) signals plays a vital role in the early detection and diagnosis of heart diseases. Analyzing ECG signals manually is a time-consuming process that requires expertise, which might cause a delay in the diagnosis and treatment of heart diseases. This project aims at developing an automatic ECG signal classification system based on the application of Deep Learning techniques. ECG signals are collected from the publicly available MIT-BIH Arrhythmia Dataset, which contains labeled ECG signals for both normal and abnormal heartbeats.

The collected ECG signals are preprocessed before training the model for classification. Deep learning techniques, such as Convolutional Neural Network (CNN) architectures, including ResNet and VGGNet, are applied for the automatic classification of ECG signals. This helps the model learn the features and classify the signals with high accuracy. This project has been implemented using the Google Colab environment with the help of the Python programming language and TensorFlow and Keras libraries. This project shows the efficiency of the application of deep learning techniques for the classification of ECG signals.

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